DC-DC Boost Converter Prototype

1. INTRODUCTION

The converter boosts the voltage from the photovoltaic (PV) panel to a regulated 36V DC output suitable for EV charging applications. This report includes the work completed during Weeks 3 and 4 of the project. The system requirements were identified and the main components were selected. Design calculations were also carried out to determine the duty cycle, inductor value, capacitor value, and load resistance required for the boost converter.

2. SYSTEM SPECIFICATIONS

Parameter	Value
PV Panel Power	100 W
Open Circuit Voltage (Voc)	22 V
Maximum Power Voltage (Vmp)	18 V
Output Voltage (Vout)	36 V
Switching Frequency	20 kHz
Converter Type	DC-DC Boost Converter

- Solar panel : 100W flexible monocrystalline
  This was slected because its physical surface area and the flexible design have around 2kg which will be fit safely on the vehicle frame
- Battery : 36v
  This was selected because the protoype vehicle needs a lot of heavy electrical power to move around. If we used a low 12V battery to get that same power, it draws current to through our thin prototype wires. This high current would create huge heat losses, causing the converter components and wires to instantly overheat and melt. But need to reconsider cause this affords much higher.

3. DESIGN CALCULATIONS

Input Voltage (Vin) = 18V
Output Voltage (Vout) = 36V
Output Power = 100W
Switching Frequency = 20 kHz

4.1 Duty Cycle

For a boost converter:

$$D = 1 - \frac{Vin}{Vout}$$

$$D = 1 - (18 / 36)$$
$$D = 0.5$$

Required Duty Cycle = 50%

4.2 Input Current

$$I_{in} = P_{out} / V_{in}$$

$$I_{in} = 100 / 18$$

$$I_{in} = 5.56 \text{ A}$$

4.3 Inductor Calculation

Assume ripple current = 30% of input current

$\Delta_{IL} = 0.3 \times 5.56$

$\Delta_{IL} = 1.67$ A

Inductor,

$$L = \frac{(Vin \times D)}{(\Delta IL \times f)}$$

$$L = \frac{(18 \times 0.5)}{(1.67 \times 20000)}$$

$$L = 269\ \mu H$$

Inductor = 220 µH – 330 µH
This helps maintain continuous conduction mode (CCM) and reduces current ripple.

<u>4.4 Output Capacitor</u>

Output current:

$$I_{out} = Pout / Vout$$

$$I_{out} = 100 / 36$$

$$I_{out} = 2.78 \text{ A}$$

Allowable ripple voltage:

$$\Delta V_{out} = 1\% \times 36$$

$$\Delta V_{out} = 0.36 \text{ V}$$

Capacitor :

$$C = \frac{(\text{Iout} \times \text{D})}{(\text{Vout} \times \text{f})}$$

$$C = \frac{(2.78 \times 0.5)}{(0.36 \times 20000)}$$

C = 193 μF

Selected value:

Capacitor = 220 μF,470 μF electrolytic capacitor

A higher capacitor value helps reduce output ripple and improve voltage stability.

4.5 Load Resistance

$$R = Vout^2 / Pout$$

$$R = 36^2 / 100$$

$$R = 12.96\ \Omega$$

Selected practical value:

Load resistor = 15 Ω high-power wirewound resistor

4.6 Switching Frequency Selection

Initially, a switching frequency of 50 kHz was considered to reduce passive component size.
However 20 kHz was selected for the prototype because it provides:

- Lower MOSFET switching losses
- Reduced heating
- Easier hardware implementation
- Better stability on breadboard/PCB
- Lower EMI noise
- Easier debugging with oscilloscope.

5. MAIN COMPONENTS SELECTED

Component	Specification
MOSFET	IRFZ44N N-Channel MOSFET
Diode	MBR20100CT Schottky Diode
Inductor	330 µH Toroidal Inductor
Capacitor	220 µF Electrolytic Capacitor
Gate Driver	TLP250 / IR2110
PWM Controller	Arduino / MCU PWM Generation